WORKED SOLUTION

STEP II 1989, Question 1

Pure • Trigonometric Solution of Cubics

STEP II 1989, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s2-q1

The key idea

The identity says that $4z^3 - 3z$ is just $\cos 3\theta$ in disguise when $z = \cos\theta$. So if we can squeeze the cubic into the shape $4z^3 - 3z = k$ with $|k| \leq 1$, solving it is the same as solving $\cos 3\theta = k$, which has three easy solutions. The two substitutions do two separate jobs: the shift $y = x + a$ removes the squared term, and the scaling $z = by$ fixes the ratio of the remaining coefficients to $4 : 3$.

Solution

The identity

Using the addition formulae and $\sin^2\theta = 1 - \cos^2\theta$,

$$\begin{aligned}\cos 3\theta &= \cos 2\theta\cos\theta - \sin 2\theta\sin\theta \\ &= (2\cos^2\theta - 1)\cos\theta - 2\sin^2\theta\cos\theta \\ &= 2\cos^3\theta - \cos\theta - 2(1 - \cos^2\theta)\cos\theta \\ &= 4\cos^3\theta - 3\cos\theta,\end{aligned}$$

as required.

Reducing the cubic

With $y = x + a$ we have $x = y - a$, and the coefficient of y^2 in $24(y-a)^3 - 72(y-a)^2 + \cdots$ is $-72a - 72$. To remove it we need $a = -1$, that is $x = y + 1$. Then

$$\begin{aligned}&24(y+1)^3 - 72(y+1)^2 + 66(y+1) - 19 \\ &\quad= (24y^3 + 72y^2 + 72y + 24) - (72y^2 + 144y + 72) \\ &\qquad + (66y + 66) - 19 \\ &\qquad\qquad = 24y^3 - 6y - 1,\end{aligned}$$

so $(*)$ becomes

$$24y^3 - 6y = 1.$$

Now put $y = z/b$:

$$\frac{24}{b^3}z^3 - \frac{6}{b}z = 1.$$

Ask yourself How do we choose b so that the left-hand side is a multiple of $4z^3 - 3z$?

We need the coefficients of z^3 and z in the ratio $4 : -3$, so

$$\frac{24/b^3}{6/b} = \frac{4}{3} \quad \Longrightarrow \quad \frac{4}{b^2} = \frac{4}{3} \quad \Longrightarrow \quad b^2 = 3.$$

Take $b = \sqrt{3}$. Then the equation is $\dfrac{8}{\sqrt{3}}z^3 - \dfrac{6}{\sqrt{3}}z = 1$, and multiplying by $\frac{\sqrt{3}}{2}$,

$$4z^3 - 3z = \frac{\sqrt{3}}{2}.$$

So $a = -1$, $b = \sqrt{3}$ and $k = \frac{\sqrt{3}}{2}$, with $z = \sqrt{3}\,(x - 1)$. (Taking $b = -\sqrt{3}$ instead gives $k = -\frac{\sqrt{3}}{2}$ and the same roots in the end.)

The roots

Since $|k| \leq 1$ we can write $z = \cos\theta$, and by the identity the equation becomes

$$\cos 3\theta = \frac{\sqrt{3}}{2} = \cos 30^\circ.$$

So $3\theta = 30^\circ,\ 390^\circ,\ 750^\circ$, giving $\theta = 10^\circ,\ 130^\circ,\ 250^\circ$. These give three different values of $\cos\theta$, and a cubic has at most three roots, so they are all of them:

$$z = \cos 10^\circ, \quad \cos 130^\circ, \quad \cos 250^\circ.$$

Undoing the substitutions, $x = 1 + \dfrac{z}{\sqrt{3}}$, so

$$x_1 = 1 + \frac{\cos 10^\circ}{\sqrt{3}} = 1.5686\ldots \approx 1.57,$$
$$x_2 = 1 + \frac{\cos 130^\circ}{\sqrt{3}} = 0.6289\ldots \approx 0.629,$$
$$x_3 = 1 + \frac{\cos 250^\circ}{\sqrt{3}} = 0.8025\ldots \approx 0.803.$$

Not every cubic works

For a general monic cubic $x^3 + \alpha x^2 + \beta x + \gamma = 0$, the shift is forced: the coefficient of y^2 after $x = y - a$ is $\alpha - 3a$, so $a = \frac{\alpha}{3}$ and we reach $y^3 + py + q = 0$ with $p = \beta - \frac{\alpha^2}{3}$. Putting $y = z/b$ and multiplying by $4b^3$ gives

$$4z^3 + 4pb^2z + 4qb^3 = 0,$$

and to match $4z^3 - 3z$ we need $4pb^2 = -3$, that is $b^2 = -\dfrac{3}{4p}$. This has a real solution only if $p < 0$.

A counterexample is

$$x^3 + 3x - 4 = 0.$$

Here $\alpha = 0$, so $a = 0$ and $p = 3 > 0$; we would need $b^2 = -\frac{1}{4}$, which no real b satisfies. So the equation cannot be put in the form $4z^3 - 3z = k$ at all. (It does have a root: it factorises as $(x-1)(x^2+x+4) = 0$, and the quadratic has discriminant $-15 < 0$, so $x = 1$ is its only real root. The method fails, not the equation.)

Even when $p < 0$ the method can fail at the next step. For $x^3 - 3x - 4 = 0$, the substitution $x = 2z$ gives $4z^3 - 3z = 2$; but $|\cos 3\theta| \leq 1$, so there is no real θ with $\cos 3\theta = 2$.

Checks

- The three roots should add to $\frac{72}{24} = 3$ and multiply to $\frac{19}{24}$. Indeed the cos terms sum to zero ($\cos 10^\circ + \cos 130^\circ + \cos 250^\circ = 0$), and numerically $1.5686 \times 0.6289 \times 0.8025 = 0.7917 \approx \frac{19}{24}$.
- Substituting $x = 1.5686$ into $24x^3 - 72x^2 + 66x - 19$ gives zero to the accuracy of a calculator; the same holds for the other two roots.

Common slips

- Stopping at $\theta = 10^\circ$: the other two roots come from $3\theta = 390^\circ$ and 750°.
- Getting the sign of a wrong: $y = x + a$ with $x = y + 1$ means $a = -1$, not $+1$.
- For the counterexample, choosing a cubic with $p < 0$ and $|k| \leq 1$, which the method handles perfectly well. Check $b^2 > 0$ and $|k| \leq 1$ before claiming failure.

Beyond STEP

The method works exactly when the cubic has three real roots, the "casus irreducibilis" in which Cardano's formula forces you through complex cube roots even though every answer is real. When $|k| > 1$ the same idea works with cosh in place of cos, since $\cosh 3u = 4\cosh^3 u - 3\cosh u$; for $x^3 - 3x - 4 = 0$ this gives the real root $x = 2\cosh\left(\frac{1}{3}\operatorname{arcosh} 2\right) \approx 2.196$. The deeper question of which polynomial equations can be solved by radicals at all leads to Galois theory; Allenby, *Rings, Fields and Groups*, is a good place to start, and Needham, *Visual Complex Analysis*, shows the geometry of the cube roots.

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